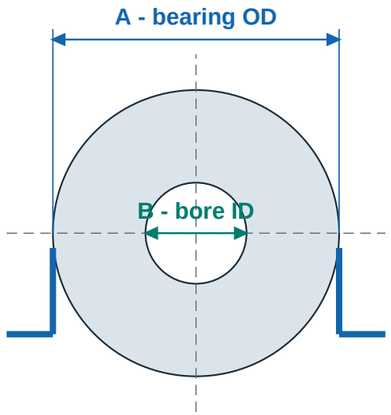
MEASURE YOUR DRAW-STUD PUNCH HARDWARE

Bearing size and stud travel - keep the bearing and its washers in their normal working order.

1 Bearing: face view

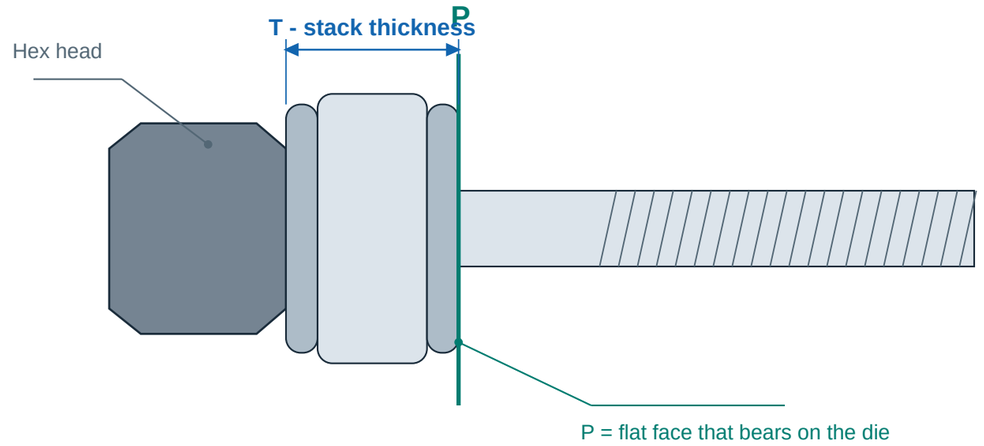


A: Outside jaws touch the two opposite outside tangencies.

B: Inside jaws touch opposite bore walls, if accessible.

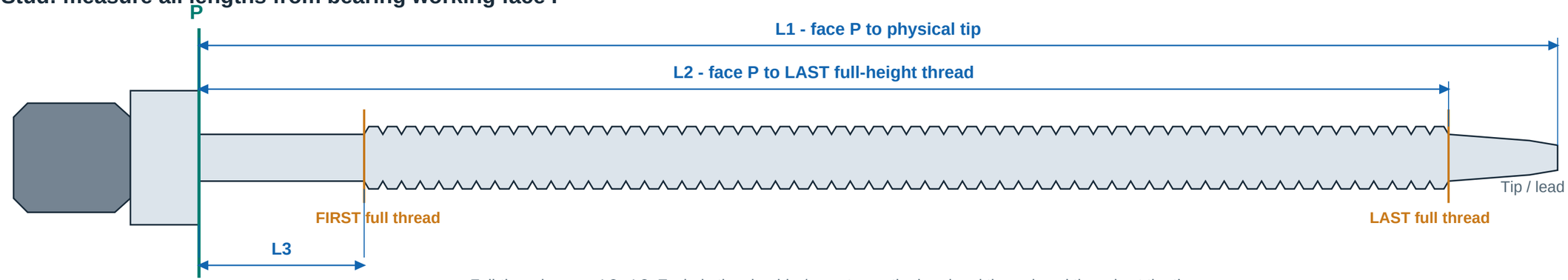
B is optional if the bearing is captive. Do not pry it off.
Reuse the complete bearing stack; its ID is not needed to cut a new die.

2 Bearing: side view / working face P



T: Include both races and any washers normally under the head.
Light finger contact only. If either end face is inaccessible, report "captive".

3 Stud: measure all lengths from bearing working face P



Full-thread span = L2 - L3. Exclude the shoulder/runout near the head and the reduced threads at the tip.
Seat the bearing lightly against the head. Use the same exposed face P for L1, L2 and L3.

SCHEMATIC - NOT TO SCALE | Smaller stud confirmed by owner: 7/16-20 UNF (1.27 mm pitch).

Record actual values in mm to 0.1 mm. These are measurements, not manufacturing tolerances or a load rating.

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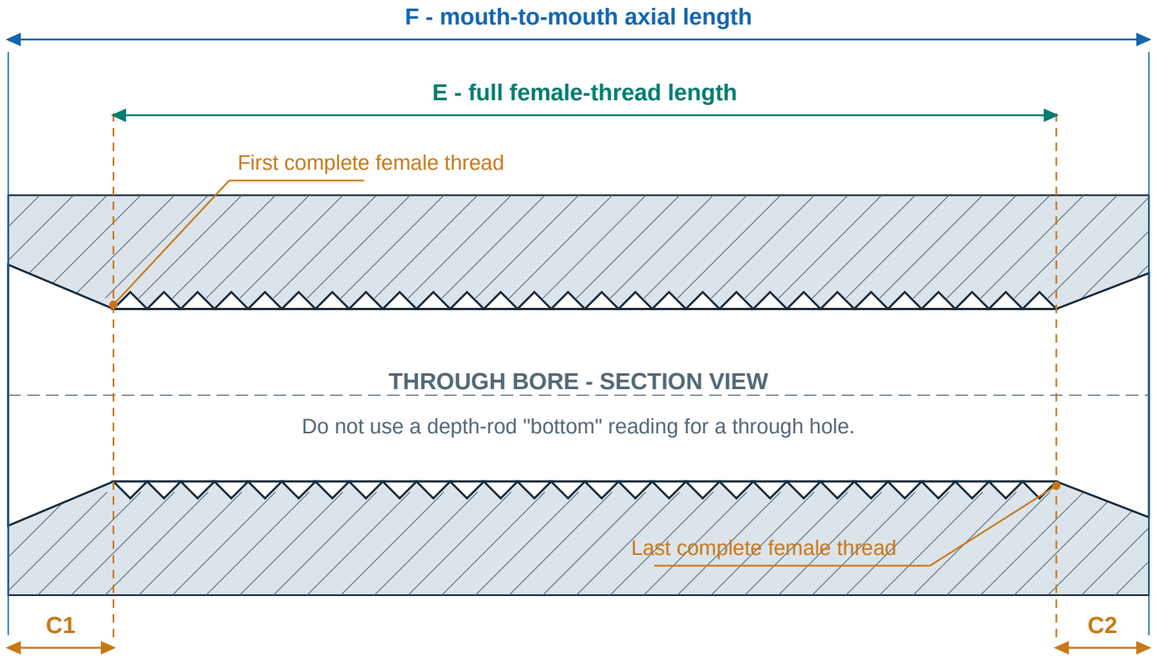
Existing punch: identify the full female-thread zone. Chamfers and incomplete threads do not count.

4 Punch: section through the threaded bore



Entry faces the die. Reference the bore mouth, not the tallest cutting tooth.

HOW TO MEASURE THE PUNCH

F: Outside jaws between the two bore-mouth planes, where those planes are accessible.

C1 / C2: From each mouth to its first complete thread; include chamfers and thread runout.

Use a magnifier. A depth rod measures these offsets only if the actual boundary is reachable.

E = F - C1 - C2. If the boundaries are unclear, give F and mark C1/C2 unknown.

If the bore is blind, report that fact; bottom depth is not the same as threaded depth.

5 Fill in and send these values

A	Bearing outside diameter	_____ mm
B	Bearing bore ID (or captive)	_____ mm
T	Bearing + washer stack thickness	_____ mm
L1	P to physical stud tip	_____ mm
L2	P to last complete male thread	_____ mm
L3	P to first complete male thread	_____ mm
F	Punch mouth-to-mouth length	_____ mm
C1	Entry chamfer + incomplete threads	_____ mm
C2	Exit chamfer + incomplete threads	_____ mm

Face P flat, or is there a projecting center boss?

If a boss is present: give its OD and projection.

For captive parts, leave inaccessible values blank.

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